Multiclass Logistic Regression and Discriminant Analysis

Week 6

Review

In the last lecture, we have learned the logistic regression for binary classification $Y \in \{0, 1\}$.

- Estimating the Bayes rule at any observation $x \in \mathcal{X}$ is equivalent to estimate the conditional probability $\mathbb{P}(Y = 1 \mid X = x)$.
- Logistic regression parametrizes the conditional probability by

$$\mathbb{P}(Y = 1 \mid X = x) = \frac{e^{\beta_0 + x^\top \beta}}{1 + e^{\beta_0 + x^\top \beta}}.$$

- We estimate the coefficients by using MLE which can be solved by (stochastic) gradient descent.

Motivating Example

Recall our motivating example, where we are trying to predict the medical condition of a patient in the emergency room on the basis of her symptoms.
There are three possible diagnoses: stroke, drug overdose, and epileptic seizure.
We define Y as follows:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

How do we extend the ideas we discussed last week to the case where we have more than two classes?

Extension to multi-class classification

When $Y \in \{0, 1, \ldots, K-1\}$ for $K > 2$, we need to estimate

$$p_k(x) := \mathbb{P}(Y = k \mid X = x), \qquad \forall\, 1 \leq k \leq K-1.$$

Logistic regression assumes

$$\begin{aligned} p_1(x) &= \frac{e^{\beta_0^{(1)} + x^\top \beta^{(1)}}}{1 + \sum_{k=1}^{K-1} e^{\beta_0^{(k)} + x^\top \beta^{(k)}}}. \\ &\vdots \\ p_{K-1}(x) &= \frac{e^{\beta_0^{(K-1)} + x^\top \beta^{(K-1)}}}{1 + \sum_{k=1}^{K-1} e^{\beta_0^{(k)} + x^\top \beta^{(k)}}} \end{aligned}$$

and $p_0(x) = 1 - p_1(x) - \cdots - p_{K-1}(x)$.
So essentially, we select a single class (wlog, the Kth class) to serve as baseline.

Classification

Equivalently, for $k = 1, \ldots, K-1$

$$\log\left(\frac{p_1(x)}{p_0(x)}\right) = \beta_0^{(1)} + \beta_1^{(1)} x_1 + \cdots + \beta_p^{(1)} x_p$$
$$\log\left(\frac{p_2(x)}{p_0(x)}\right) = \beta_0^{(2)} + \beta_1^{(2)} x_1 + \cdots + \beta_p^{(2)} x_p$$
$$\vdots$$
$$\log\left(\frac{p_{K-1}(x)}{p_0(x)}\right) = \beta_0^{(K-1)} + \beta_1^{(K-1)} x_1 + \cdots + \beta_p^{(K-1)} x_p$$

Choice of the baseline ($Y = 0$) is arbitrary.

So classification can be done immediately once $\beta^{(k)}$'s are estimated.

Interpretation of Coefficients

When we fit models with different baselines (i.e. different $Y = 0$), the coefficient estimates will differ.
However, the predictions, log odds between any pair of classes, and the other key model outputs will remain the same.
The interpretation of the coefficients must be done with care since the coefficients are tied to the choice of baseline.

Motivating Example: if we set epileptic seizure to be the baseline, then a one unit increase in x is associated with a $\beta_0^{(1)}$ increase in the log odds of stroke over epileptic seizure.

How to estimate coefficients?

A naive approach: separate binary logistic regressions

$$\log\left(\frac{p_k(x)}{p_0(x)}\right) = \beta_0^{(k)} + \beta_1^{(k)} x_1 + \cdots + \beta_p^{(k)} x_p, \qquad \forall\, 1 \le k \le K-1.$$

Partition the data into $\{\mathcal{D}_{(1)}^{\text{train}}, \ldots, \mathcal{D}_{(K-1)}^{\text{train}}\}$ with $\mathcal{D}_{(k)}^{\text{train}}$ containing all data with $y \in \{0, k\}$ for $1 \le k \le K-1$.

1. For each $1 \le k \le K-1$, use $\mathcal{D}_{(k)}^{\text{train}}$ to perform binary logistic regression to estimate $\beta^{(k)}$ and estimate $\frac{p_k(x)}{p_0(x)}$

2. Assign class label by comparing

$$1, \ \frac{p_1(x)}{p_0(x)}, \ \frac{p_2(x)}{p_0(x)} \ldots, \ \frac{p_{K-1}(x)}{p_0(x)}$$

Why naive?

- Estimation of $\beta^{(k)}$
- only uses $\mathcal{D}_{(k)}^{\text{train}}$ containing data points in class $\{0, k\}$
- ignore all data points in other classes

- The event $\{y_i = k\}$ is dependent on all other $\{y_i = k'\}$ for $k' \neq k$. Intuitively, this dependence helps to estimate $\beta^{(k)}$ by pooling data from all classes.

- What should we use instead?

MLE for multi-class logistic regression

The conditional log-likelihood of $y_1, \ldots, y_n \mid x_1, \ldots, x_n$ at $(\beta^{(1)}, \ldots, \beta^{(K-1)})$, with no intercepts, is proportional to

$$
\begin{aligned}
&\sum_{i=1}^{n} \log \left(\prod_{k=0}^{K-1} p_k(x_i)^{\mathbf{1}\{y_i = k\}} \right) \\
&= \sum_{i=1}^{n} \sum_{k=0}^{K-1} \mathbf{1}\{y_i = k\} \log(p_k(x_i)) \\
&= \sum_{i=1}^{n} \left[\mathbf{1}\{y_i = 0\} \log(p_0(x_i)) + \sum_{k=1}^{K-1} \mathbf{1}\{y_i = k\} \log(p_k(x_i)) \right] \\
&= \sum_{i=1}^{n} \left[\sum_{k=1}^{K-1} \mathbf{1}\{y_i = k\} x_i^\top \beta^{(k)} - \sum_{k=0}^{K-1} \mathbf{1}\{y_i = k\} \log \left(1 + \sum_{k=1}^{K-1} e^{x_i^\top \beta^{(k)}} \right) \right] \\
&= \sum_{i=1}^{n} \left[\sum_{k=1}^{K-1} \mathbf{1}\{y_i = k\} x_i^\top \beta^{(k)} - \log \left(1 + \sum_{k=1}^{K-1} e^{x_i^\top \beta^{(k)}} \right) \right]
\end{aligned}
$$

Gradient of $\ell(\beta^{(k)})$

For any $1 \leq k \leq K-1$,

$$\begin{aligned}\frac{\partial \ell(\beta^{(1)}, \ldots, \beta^{(K-1)})}{\partial \beta^{(k)}} &= \sum_{i=1}^{n} \left[\mathbf{1}\{y_i = k\} x_i - \frac{x_i e^{x_i^\top \beta^{(k)}}}{1 + \sum_{k=1}^{K-1} e^{x_i^\top \beta^{(k)}}} \right] \\ &= \sum_{i=1}^{n} \left[\mathbf{1}\{y_i = k\} - \frac{e^{x_i^\top \beta^{(k)}}}{1 + \sum_{k=1}^{K-1} e^{x_i^\top \beta^{(k)}}} \right] x_i\end{aligned}$$

The binary case ($K = 2$)

$$\begin{aligned}\frac{\partial \ell(\beta)}{\partial \beta} &= \sum_{i=1}^{n} \left[\mathbf{1}\{y_i = 1\} - \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}} \right] x_i \\ &= \sum_{i=1}^{n} \left[y_i - \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}} \right] x_i.\end{aligned}$$

Gradient descent

The log-likelihood we just showed is concave, thus, it's negative is convex. Therefore, for $1 \leq k \leq K$, we update

$$\hat{\beta}_{(t+1)}^{(k)} = \hat{\beta}_{(t)}^{(k)} + \alpha \sum_{i=1}^{n} \left[\mathbf{1}\{y_i = k\} - \frac{e^{x_i^\top \hat{\beta}_{(t)}^{(k)}}}{1 + \sum_{k=1}^{K-1} e^{x_i^\top \hat{\beta}_{(t)}^{(k)}}} \right] x_i.$$

Remark:

- the gradient update uses data points from **all classes**!
- better estimation than the naive approach

An alternative to Logistic Regression

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable[1].
- Discriminant analysis does not suffer from this problem.

- When n is small and we know more about the data, such as the distribution of $X \mid Y = k$
- Discriminant analysis has better performance than the logistic regression model.

- Logistic Regression sometimes does not handle multi-class classification well
- Discriminant analysis is more suitable for multi-class classification problems.

1. **A paper on this.**

Discriminant Analysis

Discriminant Analysis

Suppose we're trying to classify whether a student is an undergraduate or graduate student based on their age.

Logistic regression: Given that a student is 25 years old, what is the probability that they're a graduate student?

$$\mathbb{P}(Y = 1 \mid X = x)$$

- We know the student's **age** (X), we want to predict their **student status** (Y)
Discriminant Analysis: What does the distribution of ages look like among undergraduate students? What about among graduate students?

$$X \mid Y = k, \qquad \forall k \in \mathcal{C}.$$

- We look at the distribution of the predictors within each class.

Discriminant Analysis

When we generalize the previous example:

- Logistic regression directly parametrizes

$$\mathbb{P}(Y = k \mid X = \boldsymbol{x}), \qquad \forall k \in \mathcal{C}.$$

- By contrast, **Discriminant Analysis** parametrizes the distribution of

$$X \mid Y = k, \qquad \forall k \in \mathcal{C}.$$

- Normal distributions are oftentimes used.

Discriminant Analysis

What does parametrizing $X \mid Y = k$ buy us?

- By Bayes' theorem,

$$\mathbb{P}(Y = k \mid X = \boldsymbol{x}) = \frac{\mathbb{P}(X = \boldsymbol{x} \mid Y = k)\mathbb{P}(Y = k)}{\mathbb{P}(X = \boldsymbol{x})}.$$

- Thus, to compare two classes $k, k' \in \mathcal{C}$ with $k \neq k'$

$$\begin{aligned} &\mathbb{P}(Y = k \mid X = \boldsymbol{x}) \geq \mathbb{P}(Y = k' \mid X = \boldsymbol{x}) \\ \iff &\mathbb{P}(X = \boldsymbol{x} \mid Y = k)\mathbb{P}(Y = k) \geq \mathbb{P}(X = \boldsymbol{x} \mid Y = k')\mathbb{P}(Y = k') \end{aligned}$$

Notation for discriminant analysis

Suppose we have K classes, $\mathcal{C} = \{0, 1, 2, \ldots, K-1\}$. For any $k \in \mathcal{C}$,

- We write

$$\pi_k := \mathbb{P}(Y = k)$$

as the prior probability that a randomly chosen observation comes from the kth class.

- Write

$$f_k(x) := \mathbb{P}(X = \boldsymbol{x} \mid Y = k)$$

as the conditional density function of $X = x$ from class k.

- In discriminant analysis, parametric assumption is assumed on $f_k(\boldsymbol{x})$.

The Bayes rule

- By the Bayes' theorem,

$$p_k(x) := \mathbb{P}(Y = k \mid X = \boldsymbol{x}) = \frac{\pi_k f_k(\boldsymbol{x})}{\sum_{\ell \in \mathcal{C}} \pi_\ell f_\ell(\boldsymbol{x})}$$

is called the posterior probability, i.e. the probability that an observation belongs to the kth class given its feature.

- According to the Bayes classifier, we should classify a new point x according to

$$\arg\max_{k \in \mathcal{C}} p_k(\boldsymbol{x}) = \arg\max_{k \in \mathcal{C}} \frac{\pi_k f_k(\boldsymbol{x})}{\sum_{\ell \in \mathcal{C}} \pi_\ell f_\ell(\boldsymbol{x})} = \arg\max_{k \in \mathcal{C}} \pi_k f_k(\boldsymbol{x}).$$

Discriminant Analysis for $p = 1$

Assume that

$$X \mid Y = k \sim N(\mu_k, \sigma_k^2), \qquad \forall k \in \mathcal{C},$$

namely,

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2\sigma_k^2}(x-\mu_k)^2}.$$

Linear Discriminant Analysis (LDA) further assumes

$$\sigma_0^2 = \sigma_1^2 = \cdots = \sigma_{K-1}^2 = \sigma^2.$$

Linear Discriminant Analysis for $p = 1$

As a result,

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{\ell \in \mathcal{C}} \pi_\ell f_\ell(x)} = \frac{\pi_k e^{-\frac{1}{2\sigma^2}(x-\mu_k)^2}}{\sum_{\ell \in \mathcal{C}} \pi_\ell e^{-\frac{1}{2\sigma^2}(x-\mu_\ell)^2}}.$$

The Bayes rule classifies $X = x$ to

$$\begin{aligned} \arg\max_{k \in \mathcal{C}} p_k(x) &= \arg\max_{k \in \mathcal{C}} \log(p_k(x)) \\ &= \arg\max_{k \in \mathcal{C}} \underbrace{\frac{\mu_k}{\sigma^2} x - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k}_{\delta_k(x)} \qquad \text{(verify!)} \end{aligned}$$

The name LDA is due to the fact that the discriminant function $\delta_k(x)$ is a linear function in x.

Linear Discriminant Analysis for $p = 1$

As an example, let $K = 2$:

$$\underset{k\in\{0,1\}}{\arg\max} p_k(x) = \underset{k\in\{0,1\}}{\arg\max} \left[\frac{\mu_k}{\sigma^2} x - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k \right]$$

In this example, let's also allow the priors are equal $\pi_0 = \pi_1$ and suppose $\mu_1 \geq \mu_0$. Then the Bayes classifier assigns $X = x$ to

$$\begin{cases} 0 & \text{if } x < \frac{\mu_0+\mu_1}{2} \\ 1 & \text{if } x \geq \frac{\mu_0+\mu_1}{2} \end{cases}$$

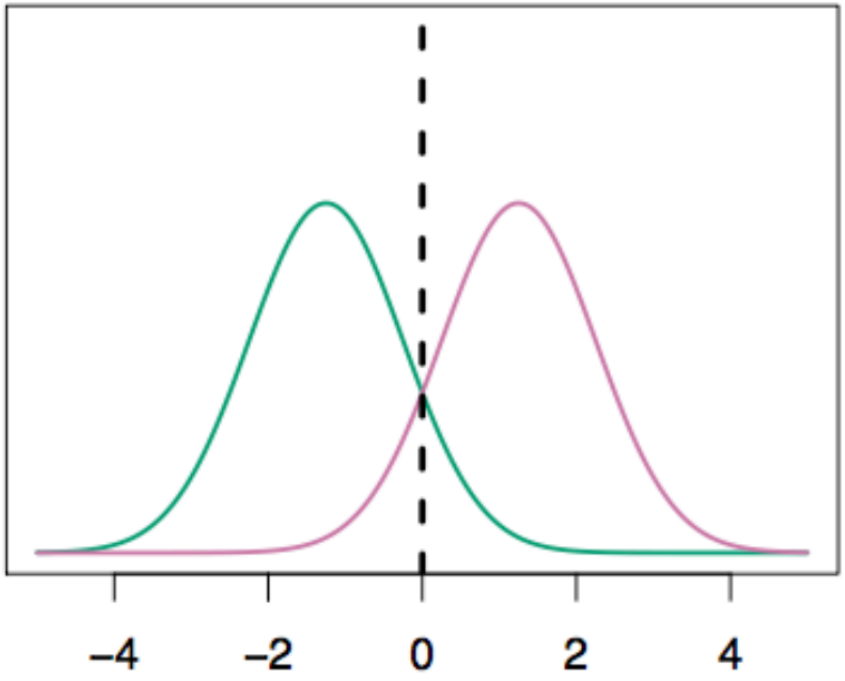
The line $x = (\mu_0 + \mu_1)/2$ is called the **Bayes decision boundary**.

Example of LDA in binary classification

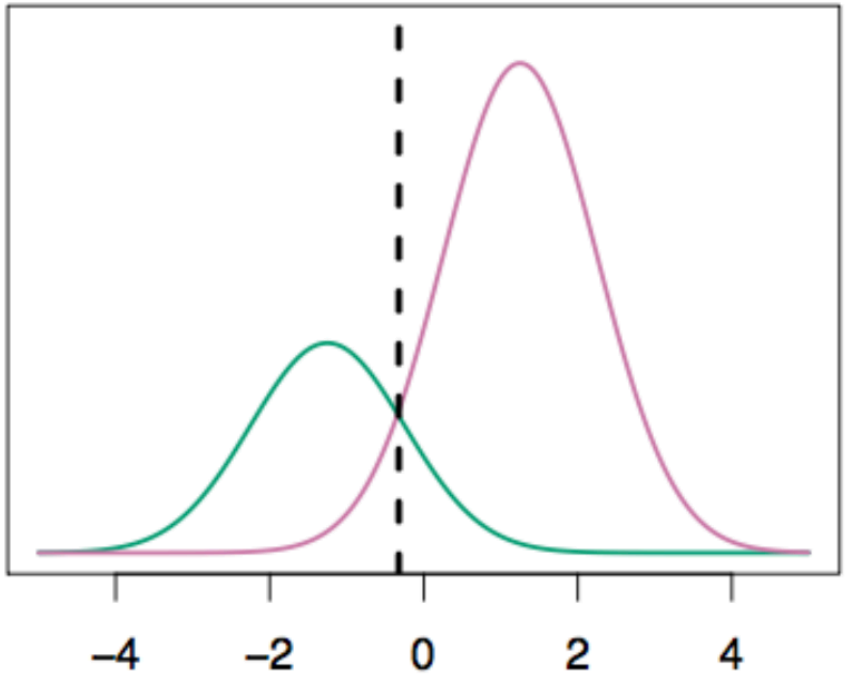
Consider $\mu_0 = -1.5$, $\mu_1 = 1.5$, and $\sigma = 1$. The curves are $p_0(x)$ (green) and $p_1(x)$ (red). The dashed vertical lines are the Bayes decision boundary.

$$f^*(x) = \begin{cases} 0 & \text{if } x < \frac{\mu_0+\mu_1}{2} = 0 \\ 1 & \text{if } x \geq \frac{\mu_0+\mu_1}{2} = 0 \end{cases}$$

π_1=.5, π_2=.5



π_1=.3, π_2=.7



Compute the Bayes classifier

If we know $\mu_0, \ldots, \mu_{K-1}$, σ^2 and $\pi_0, \ldots, \pi_{K-1}$, then we can construct the Bayes rule

$$\arg\max_{k \in \mathcal{C}} \delta_k(x) = \arg\max_{k \in \mathcal{C}} \left\{ \frac{\mu_k}{\sigma^2} x - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k \right\}.$$

However, we typically don't know these parameters. We need to use the training data to estimate them!

Estimation under LDA

Given training data $(x_1, y_1), \ldots, (x_n, y_n)$, for all $k \in \mathcal{C}$,
We have

$$n_k = \sum_{i=1}^{n} \mathbf{1}\{y_i = k\}.$$

We estimate π_k by $\hat{\pi}_k = \frac{n_k}{n}$.
We estimate μ_k and σ^2 by

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{1 \le i \le n : y_i = k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^{K} \sum_{1 \le i \le n : y_i = k} (x_i - \hat{\mu}_k)^2.$$

- These are actually the MLEs (verify!).

The LDA classifier

We estimate $\delta_k(x)$ by the plug-in estimator

$$\hat{\delta}_k(x) = \frac{\hat{\mu}_k}{\hat{\sigma}^2}x - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log \hat{\pi}_k.$$

The LDA classifier assigns x to

$$\arg\max_{k \in \mathcal{C}} \hat{\delta}_k(x).$$

How about the case when $p > 1$?

Linear Discriminant Analysis for $p > 1$

Recall that the posterior probability has the form

$$\mathbb{P}(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell \in \mathcal{C}} \pi_\ell f_\ell(x)},$$

Now, we assume

$$X \mid Y = k \sim N_p(\mu_k, \Sigma), \qquad \forall k \in \mathcal{C},$$

that is,

$$f_k(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^\top \Sigma^{-1}(x-\mu_k)}.$$

Linear Discriminant Analysis for $p > 1$

The discriminant function becomes

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k$$

compared to the univariate case

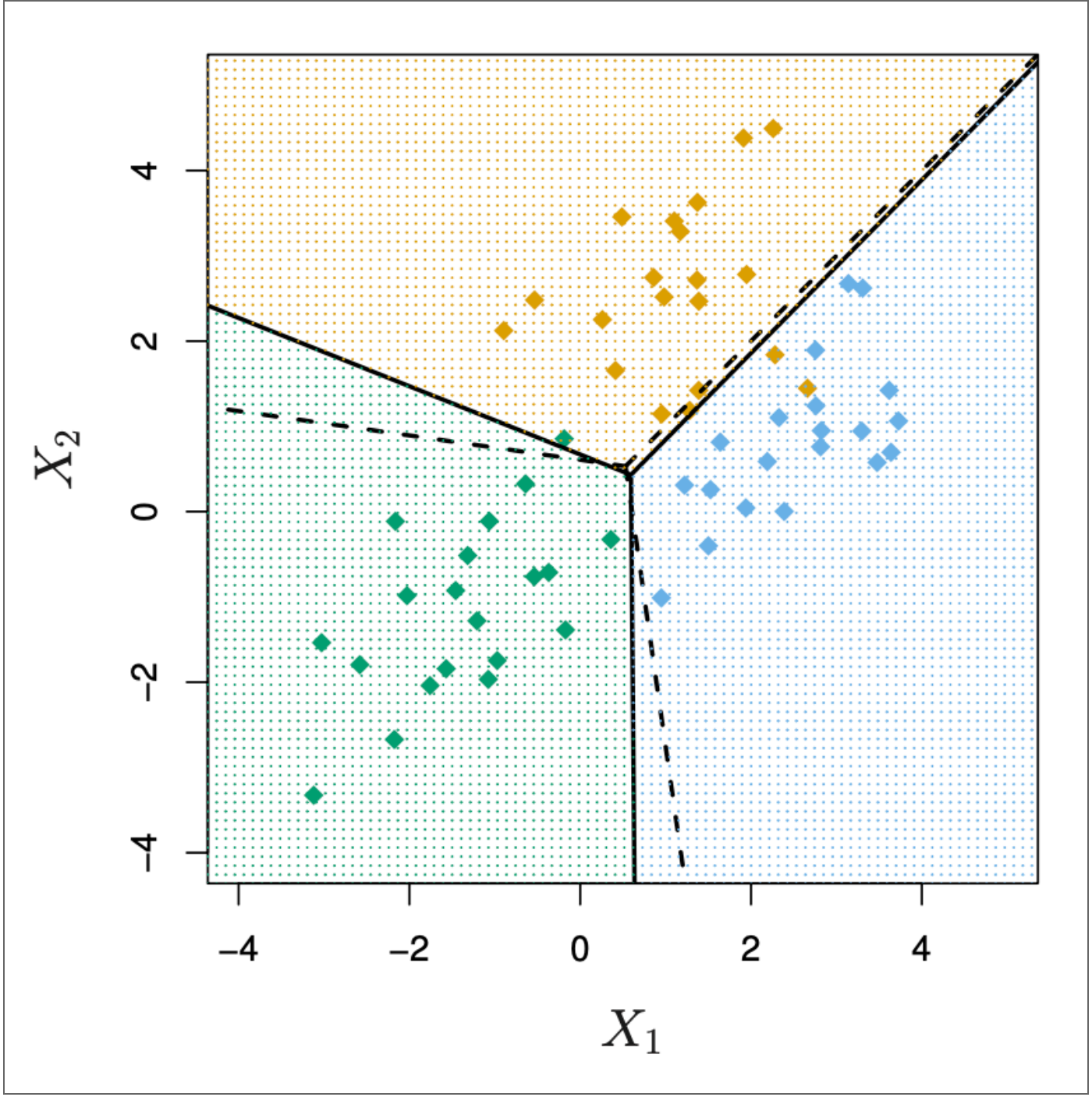
$$\delta_k(x) = \frac{\mu_k}{\sigma^2} x - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

The Bayes decision boundaries are the set of x for which

$$\delta_k(x) = \delta_\ell(x), \qquad \forall k \neq \ell,$$

which are again affine hyperplanes in $\mathbb{R}^p$.

Example



There are three classes (orange, green and blue) with two features X_1 and X_2. Dashed lines are the Bayes decision boundaries. Solid lines are their estimates based

on the LDA.

Estimation under LDA for $p > 1$

Given the training data $(x_1, y_1), \ldots, (x_n, y_n)$, for any $k \in \mathcal{C}$,
We have

$$n_k = \sum_{i=1}^{n} \mathbf{1}\{y_i = k\}.$$

We estimate π_k by $\hat{\pi}_k = \frac{n_k}{n}$.
The slight difference is to estimate μ_k and Σ by

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{1 \le i \le n : y_i = k} x_i$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^{K} \sum_{1 \le i \le n : y_i = k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^\top.$$

A plugin rule for estimating discriminant functions

We use the plugin estimator

$$\hat{\delta}_k(x) = x^\top \hat{\Sigma}^{-1} \hat{\mu}_k - \frac{1}{2} \hat{\mu}_k^\top \hat{\Sigma}^{-1} \hat{\mu}_k + \log \hat{\pi}_k, \qquad \forall k \in \mathcal{C}.$$

The resulting LDA classifier is

$$\arg\max_{k \in \mathcal{C}} \hat{\delta}_k(x).$$

One can also estimate $p_k(x)$ for each $k \in \mathcal{C}$ (using Bayes theorem as we discussed earlier in the lecture - verify!).

Logistic Regression v.s. LDA: similarity

For binary classification of LDA , one can show that

$$\begin{aligned}\log\left(\frac{p_1(x)}{1-p_1(x)}\right) &= \log\left(\frac{p_1(x)}{p_0(x)}\right) \\ &= c_0 + c_1 x_1 + \cdots + c_p x_p,\end{aligned}$$

where the $c_0, c_1, \ldots, c_p$ depends on $\pi_0, \pi_1, \mu_0, \mu_1$ and Σ.

The log-odds under LDA is also a linear form in both the parameters and the features.

Logistic Regression v.s. LDA: differences

1. LDA makes more assumption by specifying $X \mid Y$.

2. The parameters are estimated differently.

- Logistic regression uses the conditional likelihood based on $\mathbb{P}(Y \mid X)$ (known as discriminative learning).
- LDA uses the full likelihood based on $\mathbb{P}(X, Y)$ (known as generative learning).

3. If classes are well-separated, then logistic regression is not advocated.

Other forms of Discriminant Analysis

LDA specifies

$$X \mid Y = k \sim N(\mu_k, \Sigma), \qquad \forall k \in \mathcal{C}.$$

Other discriminant analyses change the specifications for $X \mid Y = k$.

Quadratic discriminant analysis (QDA) assumes

$$X \mid Y = k \sim N(\mu_k, \Sigma_k), \qquad \forall k \in \mathcal{C},$$

by allowing different Σ_k across all classes.

Quadratic Discriminant Analysis: $p = 1$

Assume that

$$X \mid Y = k \sim N(\mu_k, \sigma_k^2), \qquad \forall k \in \mathcal{C},$$

namely,

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2\sigma_k^2}(x-\mu_k)^2}.$$

As a result,

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{\ell \in \mathcal{C}} \pi_\ell f_\ell(x)} = \frac{\frac{\pi_k}{\sigma_k} e^{-\frac{1}{2\sigma_k^2}(x-\mu_k)^2}}{\sum_{\ell \in \mathcal{C}} \frac{\pi_\ell}{\sigma_\ell} e^{-\frac{1}{2\sigma_\ell^2}(x-\mu_\ell)^2}}.$$

Decision boundary of QDA

The Bayes rule classifies $X = x$ to

$$\begin{aligned}
\arg\max_{k \in \mathcal{C}} p_k(x) &= \arg\max_{k \in \mathcal{C}} \log(p_k(x)) \\
&= \arg\max_{k \in \mathcal{C}} \log\left[\frac{\pi_k}{\sigma_k} e^{-\frac{1}{2\sigma_k^2}(x-\mu_k)^2}\right] \\
&= \arg\max_{k \in \mathcal{C}} \underbrace{-\frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2} x - \frac{\mu_k^2}{2\sigma_k^2} + \log \pi_k - \log(\sigma_k)}_{\delta_k(x)}
\end{aligned}$$

The name QDA is due to the fact that $\delta_k(x)$ is quadratic in x.

Quadratic Discriminant Analysis: $p \geq 1$

$$X \mid Y = k \sim N_p(\mu_k, \Sigma_k)$$

The discriminant function becomes

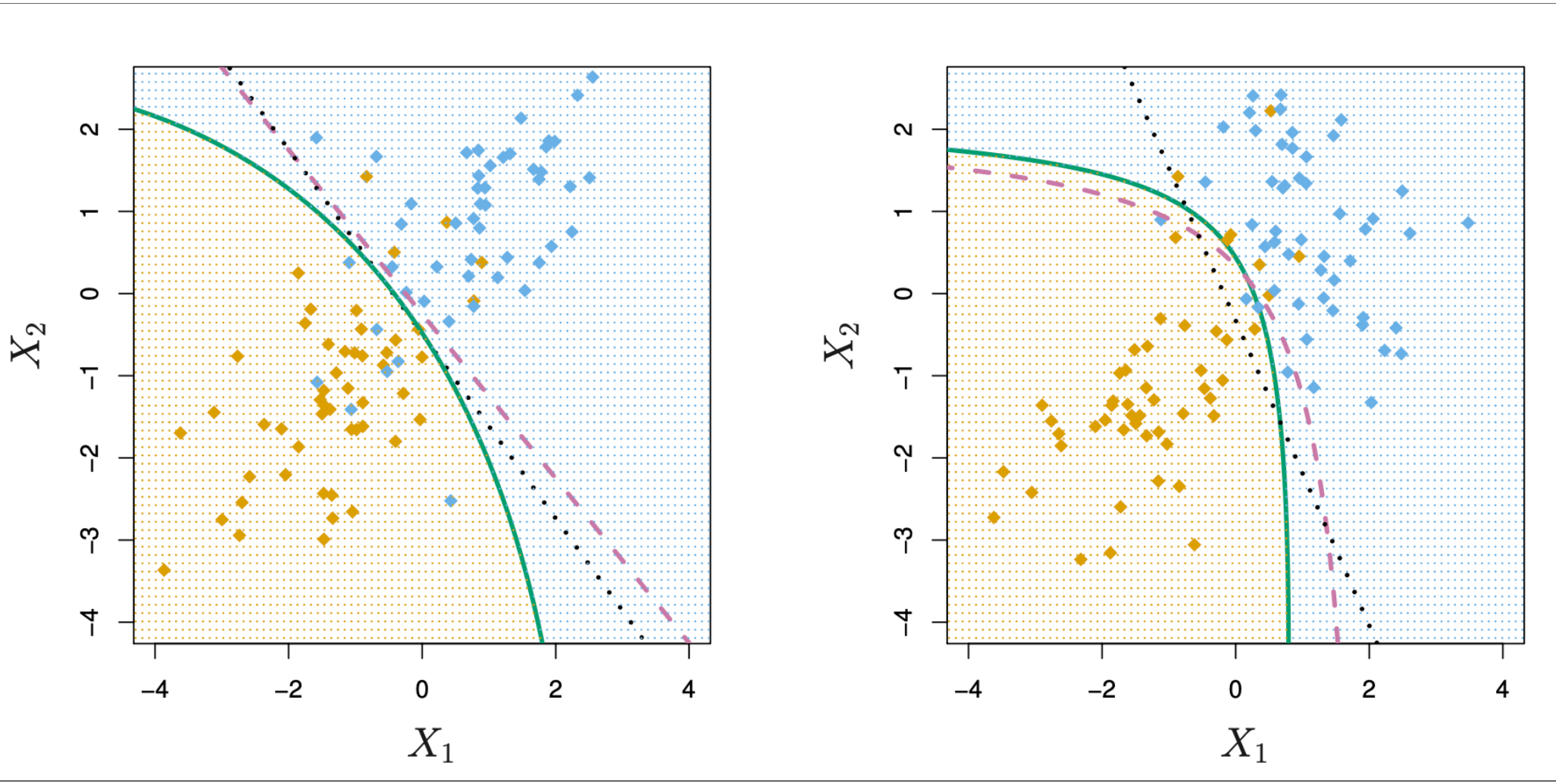
$$\begin{aligned}\delta_k(x) &= \log\left[\frac{\pi_k}{|\Sigma_k|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k)}\right] \\ &= x^\top \Sigma_k^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma_k^{-1}\mu_k + \log \pi_k - \frac{1}{2}x^\top \Sigma_k^{-1} x - \frac{1}{2}\log|\Sigma_k|.\end{aligned}$$

The decision boundary between any class k and class ℓ

$$\{x \in \mathbb{R}^p : \delta_k(x) = \delta_\ell(x)\}$$

is quadratic in x

Decision boundaries of LDA and QDA



Decision boundaries of the Bayes classifier (purple dashed), LDA (black dotted), and QDA (green solid) in two scenarios.

Estimation of QDA

Given training data $(x_1, y_1), \ldots, (x_n, y_n)$, for any $k \in \mathcal{C}$,
We have

$$n_k = \sum_{i=1}^{n} \mathbf{1}\{y_i = k\}.$$

We estimate π_k by $\hat{\pi}_k = \frac{n_k}{n}$.
We estimate μ_k and Σ_k by

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{1 \le i \le n : y_i = k} x_i$$

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{1 \le i \le n : y_i = k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^\top.$$

Plugin estimator for $\delta(x)$.

Potential problems for LDA and QDA in high dimension

LDA: we have

$$(K-1)+pK+\frac{p(p+1)}{2}$$

number of parameters to estimate.

QDA: we have

$$(K-1)+pK+\frac{p(p+1)}{2}K$$

number of parameters to estimate.

The estimation error is large when p is large comparing to n.

Naive Bayes

Other discriminant analyses: different density of $X \mid Y = k$, including non-parametric approaches.
Instead of assuming the $X \mid Y = k$ follow a family of particular distributions, **Naive Bayes** assumes

$$X_1, \ldots, X_p \text{ are independent given } Y = k$$

so that

$$f_k(x) = \prod_{j=1}^{p} f_{k,j}(x_j)$$

- It is easy to deal with both quantitative and categorical features.
- Despite the strong independence assumption within class, naive Bayes often produces good classification results.

Naive Bayes

For Gaussian density

$$X_j \mid Y = k \sim N(\mu_{k,j}, \sigma^2_{k,j}),$$

this means that $\Sigma_k = \mathrm{diag}(\sigma^2_{k,1}, \sigma^2_{k,2}, \ldots, \sigma^2_{k,p})$ and

$$f_k(x) = \prod_{j=1}^{p} \frac{1}{\sigma_{k,j}\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2_{k,j}}(x_j - \mu_{k,j})^2}$$

The discriminant function is

$$\delta_k(x) = -\frac{1}{2}\sum_{j=1}^{p} \frac{(x_j - \mu_{kj})^2}{\sigma^2_{kj}} + \log \pi_k - \frac{1}{2}\sum_{j=1}^{p} \log \sigma^2_{kj}.$$

- Derive the MLE by yourself!

Summary

- **Multiclass logistic regression** extends binary logistic regression by modelling the log-odds of each class relative to a chosen baseline class.
- **Discriminant analysis** takes a different approach: instead of modelling $Y \mid X$ directly, it models the distribution of $X \mid Y = k$ and then uses Bayes' theorem for classification.
- **LDA** assumes the classes have different means but share the same covariance matrix, leading to **linear decision boundaries**.
- **QDA** allows each class to have its own covariance matrix, giving more flexibility and **quadratic decision boundaries**.
- **Naive Bayes** simplifies the modelling of $X \mid Y$ by assuming the predictors are conditionally independent within each class.

The best classification method depends on the structure of the data and the assumptions we are willing to make.

Next Class: Tree based methods

STA314: Statistical Methods for Machine Learning I

Multiclass Logistic Regression and Discriminant Analysis

Speaker notes